



Tribhuvan University
Faculty of Humanities and Social Science

“ThriftZaar Nepal”

A PROJECT REPORT

Submitted to
Department of Computer Application
Bhaktapur Multiple Campus

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by
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Under the Supervision of
Supervisor Name

2026



Tribhuvan University
Faculty of Humanities and Social Science
Bhaktapur Multiple Campus

SUPERVISOR’S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by “name” entitled “**ThriftZaar Nepal**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

.....

SIGNATURE

<<Name>>

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Tribhuvan University
Faculty of Humanities and Social Science
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LETTER OF APPROVAL

This is to certify that this project prepared by “Kusum Darlami and Sujan Tamang” entitled “**ThriftZaar Nepal**” in partial fulfillment of the requirements for the degree of Bachelor’s in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

..... Name Supervisor Dudpati, Bhaktapur	 Mr. Madan Nath Coordinator Dudpati, Bhaktapur
..... Internal Examiner	 External Examiner

ABSTRACT

This report is submitted in partial fulfillment of the requirements for the Bachelor in Computer Application (BCA) degree at Bhaktapur Multiple Campus. It presents the design and development of “**ThriftZaar Nepal,**” a web-based platform that facilitates the online buying and selling of second-hand clothing. The system provides users with an easy and convenient way to browse, purchase, and sell pre-owned fashion items through a single digital marketplace.

The main objective of this project is to create an effective and user-friendly platform that promotes sustainable fashion while offering individuals an opportunity to generate income by selling unused clothing. The application aims to simplify the process of online transactions, enhance accessibility, and provide an organized system for managing clothing products and users.

The system is developed using **PHP, Apache, and MySQL** for backend development, while **HTML5, CSS3, and JavaScript** are used for frontend design and implementation. The website ensures efficient management of product listings, user interactions, and online transactions.

Keywords: ThriftZaar Nepal, Second-Hand Clothing, Online Marketplace, Buy and Sell Platform, Web-Based Application.

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Furthermore, I would like to express my heartfelt thanks to all respected faculty members and teachers for their academic guidance and motivation during the course of this project work. Finally, I am deeply grateful to my parents and friends for their unwavering support, encouragement, and cooperation, which enabled me to complete this project within the stipulated timeframe.

Table of Contents

SUPERVISOR’S RECOMMENDATION	i
LETTER OF APPROVAL	ii
ABSTRACT	iii
ACKNOWLEDGEMENT	iv
List Of Abbreviations	vii
List Of Figures	viii
List Of Tables	ix
Chapter 1: Introduction	1
1.1 Introduction	1
1.2 Problem Statement	2
1.3 Objectives.....	3
1.4 Scope and Limitation	3
1.4.1 Scope.....	3
1.4.2 Limitations	3
1.5 Report Organization	3
Chapter 2: Background Study and Literature Review	5
2.1 Background Study.....	5
2.2 Literature Review.....	5
Chapter 3: System Analysis and Design	9
3.1. System Analysis	9
3.1.1 Requirement Analysis	9
3.1.2 Feasibility Analysis	12
3.1.3 Data Modeling (ER-Diagram)	14
3.1.4 Process Modeling (DFD)	15
3.2 System Design.....	17
3.2.1 Architectural Design	17
3.2.2 Database Schema Design	19
3.2.3 Interface Design (UI Interface / Interface Structure Diagrams).....	20
3.2.4 Physical DFD	26
Chapter 4: Implementation and Testing	28
4.1 Implementation	28
4.1.1 Tools Used.....	28

4.1.2 Implementation Details of Modules.....	29
4.2 Testing.....	30
4.2.1 Test Cases for Unit Testing	30
4.2.2 Test Cases for System Testing.....	33
Chapter 5: Conclusion and Future Recommendations.....	34
5.1. Lesson Learnt / Outcome	34
5.2 Conclusion	34
5.3 Future Recommendations	35
Appendices.....	36
References	43

List Of Abbreviations

BCA	Bachelor Of Computer Applications
CSS	Cascading Style Sheet
DFD	Data Flow Diagram
ER	Entity Relationship
ES	ECMA Script
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
IDE	Integrated Development Environment
RDBMS	Relational Database Management System
SQL	Structured Query Language
UI	User Interface

List Of Figures

Figure 2.1: depop home page	6
Figure 2.2: depop shop by price page	6
Figure 2.3: Thrift Store Nepal home page	7
Figure 2.4: Thrift Store Nepal category page.....	8
Figure 3.1.1:Use case diagram of ThriftZaar Nepal.....	11
Figure 3.2.3.1: Home Page Interface.....	20
Figure 3.2.3.2: Login and Registration Interface	21
Figure 3.2.3.3: Product Listing Interface	22
Figure 3.2.3.4: Product Detail Interface.....	23
Figure 3.2.3.5: User Dashboard Interface	24
Figure 3.2.3.6: Admin Dashboard Interface.....	25
Figure 3.2.4.1: Physical DFD of ThriftZaar Nepal (User).....	26
Figure 3.2.4.2: Physical DFD of ThriftZaar Nepal (Admin)	27
Figure A.1: Product Listing Page.....	36
Figure A.2: Cart Page.....	36
Figure A.3: Admin Order Listing Page	37
Figure A.4: Checkout Page	37
Figure A.5: User Registration and Login Page	38
Figure A.6: Client-Side Validation.....	38
Figure A.7: Successful Registration.....	39
Figure A.8: Order Completion Page	39
Figure B.1: Database Configuration.....	40
Figure B.2: admin-remove.php	40
Figure B.3: register-logic.php	41
Figure B.4: cart-process-remove.php.....	42
Figure B.5: cancel-order.php.....	42

List Of Tables

Table 1: Test Case For User Login.....	30
Table 2: Test Case For User Registration.....	31
Table 3: Test Case For Adding and Searching Products	32
Table 4: Test Case For Adding Category	32
Table 5: Test Cases For System Testing.....	33

Chapter 1: Introduction

1.1 Introduction

ThriftZaar Nepal is a web-based platform designed to facilitate the buying and selling of second-hand clothing through an online marketplace. The system allows users to browse various categories of clothing, purchase items online, and sell their own pre-owned clothes easily. Users can register an account, log into the system, view available products, and manage their clothing listings efficiently. The platform serves as a convenient and effective medium for conducting online transactions related to second-hand fashion.

In recent years, thrift stores have evolved beyond simple resale outlets and have become important contributors to sustainable living and ethical consumption. By promoting the reuse of clothing items, the platform helps reduce environmental waste while providing users with access to affordable fashion options. ThriftZaar Nepal aims to support individuals who want to buy and sell clothing at reasonable prices through a simple and user-friendly online system.

With the increasing growth of internet usage and e-commerce services, many people prefer accessing services online rather than visiting physical stores or shopping malls. Online platforms help reduce transportation costs, minimize time consumption, avoid traffic, and provide a more convenient shopping experience from home. The availability of modern web technologies has made it possible to develop secure and reliable systems that support online transactions effectively.

The primary focus of this project is to create a centralized platform where users can easily access second-hand clothing suitable for various occasions and daily needs. Fashion trends change frequently, and individuals often require different styles of clothing. ThriftZaar Nepal provides users with an opportunity to explore a wide range of affordable clothing options with just a few clicks, ensuring convenience, accessibility, and sustainability in the modern digital environment.

1.2 Problem Statement

In recent years, the demand for second-hand clothing has increased significantly due to growing awareness of sustainable fashion and the need for affordable clothing options. Although several physical stores sell second-hand garments, customers often face challenges such as limited accessibility, high pricing of designer items, and restricted product availability. Physical thrift stores incur operational expenses such as rent, staffing, and maintenance costs, which may increase product prices and reduce affordability for many users.

Additionally, most second-hand clothing stores are concentrated in busy urban areas such as **New Road, Maitighar, Baneshwor**, and similar commercial locations. Customers living in remote or less-developed regions often find it difficult to access these stores due to travel distance, transportation costs, and time constraints. As a result, many potential buyers have limited opportunities to explore affordable and diverse clothing options.

Similarly, individuals who wish to sell their unused clothes encounter several difficulties. Traditional face-to-face selling involves time-consuming negotiations and limited exposure to potential buyers. Sellers often struggle to reach a wider audience and find suitable customers, which reduces the efficiency of the selling process. Furthermore, the absence of a centralized digital platform makes it challenging for both buyers and sellers to connect effectively.

Therefore, there is a need for a structured, accessible, and user-friendly online system that allows individuals to buy and sell second-hand clothing conveniently. The ThriftZaar Nepal project aims to address these issues by providing a centralized web-based platform where users can easily browse, purchase, and sell second-hand clothing from the comfort of their homes, thereby expanding access and improving the overall buying and selling experience.

1.3 Objectives

The main objective of this project is to develop an online system for buying and selling second-hand clothing.

As a secondary objective,

- To provide secure user registration and authentication.
- To implement search and filtering functionality.
- To design an admin dashboard for product and user management.
- To create a responsive and user-friendly interface.

1.4 Scope and Limitation

1.4.1 Scope

- Online clothing marketplace.
- User and Admin modules.
- User friendly environment.
- Time efficient.
- Project listing management.

1.4.2 Limitations

Some of the limitations that exist on our site are,

- Limited to web-based platforms.
- Users need to log in to buy clothes.
- No online payment.
- No AI based recommendations.

1.5 Report Organization

This project report is organized into five chapters, each covering distinct phases of the system development process. Chapter One presents the introduction to the project, including the background, problem statement, objectives, scope and limitations, and overall organization of the report. Chapter Two provides background study and literature review, describing fundamental concepts related to the project and an overview of existing systems along with their advantages and limitations. Chapter Three focuses on the system

analysis and design phase. It includes requirement analysis, feasibility analysis, data modeling, process modeling, and detailed system design components. Chapter Four describes the implementation and testing of the system. It includes the technologies used, module development, system functionality, and testing procedures conducted to ensure system reliability and performance. Chapter Five presents the conclusions drawn from the project, lessons learned during development, and recommendations for future improvements and enhancements of the system.

Chapter 2: Background Study and Literature Review

2.1 Background Study

With the rapid growth of internet usage and the increasing popularity of online shopping, people are now more inclined toward digital platforms for buying and selling products. At the same time, awareness of sustainable fashion and the reuse of clothing has grown significantly, leading to a rising demand for second-hand apparel. However, many existing thrift stores are limited to physical locations, making it difficult for users to access affordable clothing conveniently.

ThriftZaar Nepal was developed to address these challenges by providing an online platform where individuals can easily buy and sell second-hand clothing from the comfort of their homes. The system aims to combine the advantages of e-commerce technology with sustainable fashion practices, creating a convenient, accessible, and environmentally friendly marketplace for users.

2.2 Literature Review

Several online platforms currently provide services for buying and selling second-hand clothing. However, many existing systems present challenges such as complex navigation, poor user experience, and inefficient product browsing features. These limitations often make it difficult for users to explore available items and complete transactions smoothly. As a result, users may feel discouraged from using such platforms despite the growing demand for affordable and sustainable fashion options.

The primary objective of the ThriftZaar Nepal project is to develop a faster, more efficient, and user-friendly web application that addresses the limitations observed in existing systems. Unlike many platforms that focus on a wide range of product categories, this system is specifically designed to concentrate on second-hand and affordable clothing, providing a more organized and focused user experience.

In order to gain insights and understand best practices for implementing the system, several existing online platforms offering similar services were reviewed. The analysis of these systems, including their features, strengths, and limitations, is presented in the following sections.

depop

Depop is a mobile-focused platform popular among younger users for buying and selling vintage and second-hand clothing. It combines social media elements with an e-commerce system. Users can follow sellers, like items, and explore trending fashion styles. [1]

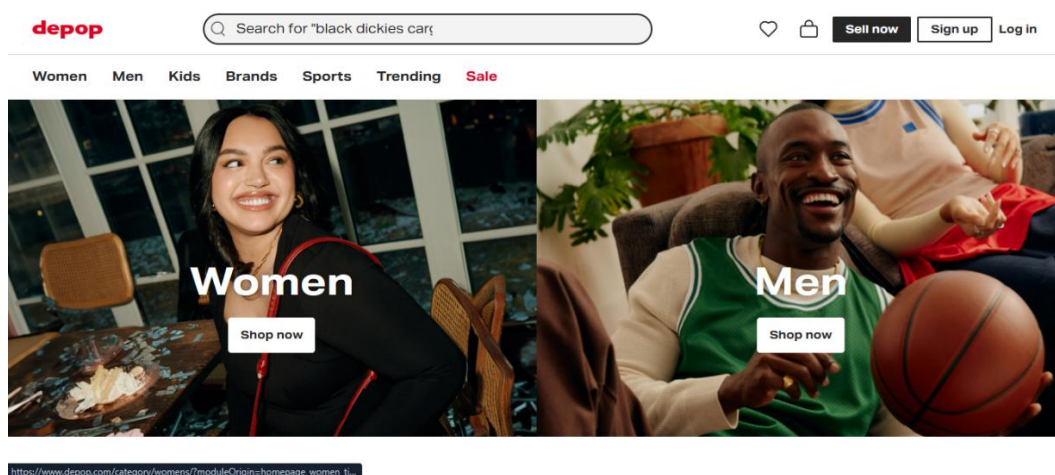


Figure 2.1: depop home page

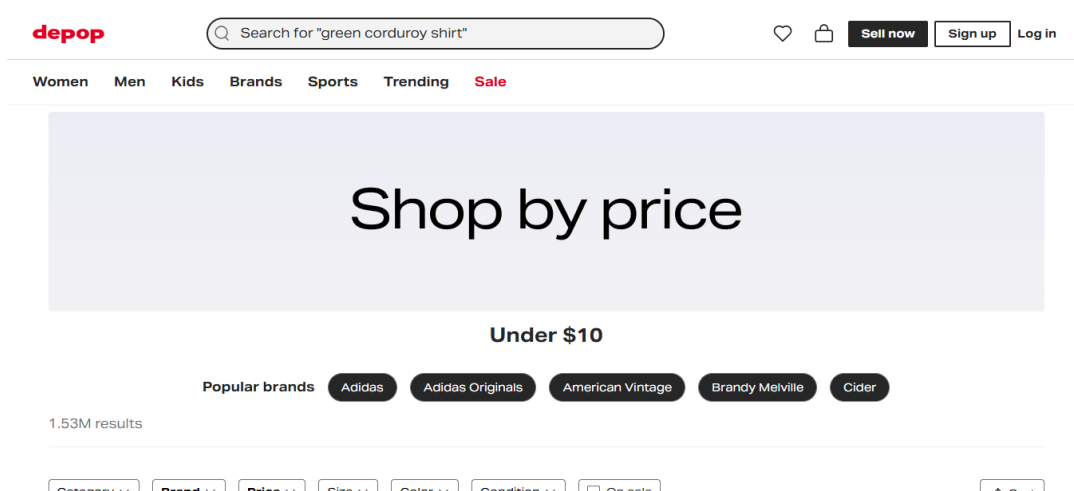


Figure 2.2: depop shop by price page

Advantages:

- Attractive visual interface
- Youth-oriented design
- Easy product listing process

Limitations:

- Primarily designed for mobile platforms
- Lack of strong administrative control
- May distract users with excessive social features

Thrift Store Nepal

Thrift Store Nepal is an online platform that promotes the buying and selling of second-hand clothing within the local Nepalese market. The website allows users to browse available thrift items and explore affordable fashion options without visiting physical stores. [2]

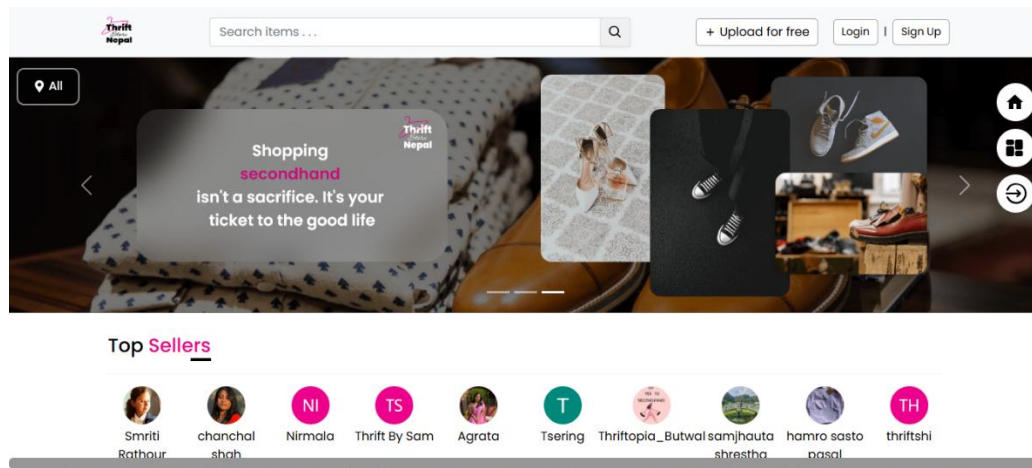


Figure 2.3: Thrift Store Nepal home page

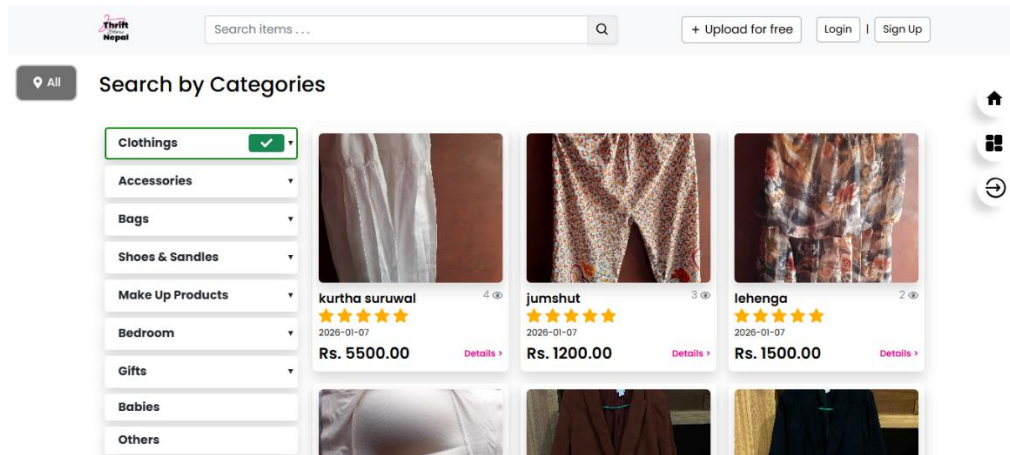


Figure 2.4: Thrift Store Nepal category page

Advantages:

- Local platform focused on Nepalese thrift fashion,
- Can upload clothes for free,
- Easy access to second-hand clothing without physical store visits,
- Supports affordable fashion for local users.

Limitations:

- No refund in case of damaged product,
- User interface and navigation can be improved,
- Lack of structured system features compared to modern e-commerce platforms,
- Limited automation in buying and selling processes,
- No proper customer support.

Chapter 3: System Analysis and Design

3.1. System Analysis

System analysis is an important phase carried out before the development of any new system. It involves proper planning of how the system will be designed, developed, tested, implemented, and maintained. [3] Effective system analysis helps in identifying project requirements, defining system objectives, and ensuring smooth workflow throughout the development process. For the successful completion of the project, all team members actively participated in discussions to analyze requirements and to plan the development strategy and overall project workflow.

3.1.1 Requirement Analysis

Requirement analysis is a crucial phase in the system development life cycle, conducted before the implementation of the system. It involves identifying, analyzing, and documenting the needs and expectations of users and stakeholders to ensure that the developed system meets its intended objectives. Proper requirement analysis helps in understanding the functionality of the system, defining system boundaries, and minimizing potential development risks.

For the development of the ThirftZaar system, requirements are mainly categorized into two major groups:

1. Functional Requirements
2. Non-Functional Requirements

Functional requirements describe the specific operations and services that the system must provide. These requirements represent the user-visible features and interactions within the system. [3] Examples include user registration, login functionality, product listing, order processing, and report generation. Functional requirements are usually defined based on the needs and expectations of stakeholders and end users.

On the other hand, non-functional requirements define how the system performs its functions and the quality attributes it must maintain. These requirements include aspects such as usability, reliability, performance, security, availability, and system implementation

constraints. Non-functional requirements ensure that the system operates efficiently, provides good user experience, and maintains data integrity and security.

Both functional and non-functional requirements are essential for designing a reliable and effective system that meets user expectations and supports the overall objectives of the ThirtZaar platform.

i. Functional Requirements

- **Administrator**
 - Manage user accounts,
 - Manage products listing,
 - Add/Edit/Delete product categories,
 - Upload/Edit/Delete clothing items,
 - Remove invalid products,
 - Login to admin dashboard.
- **Registered User (Buyer)**
 - Login and logout from the system,
 - Browse available products,
 - Place orders for clothing items,
 - Search and filter clothing items,
 - View detailed product information,
- **Guest**
 - View homepage,
 - Browse clothing products,
 - View product details,
 - Search available clothing items,
 - Register a new account.

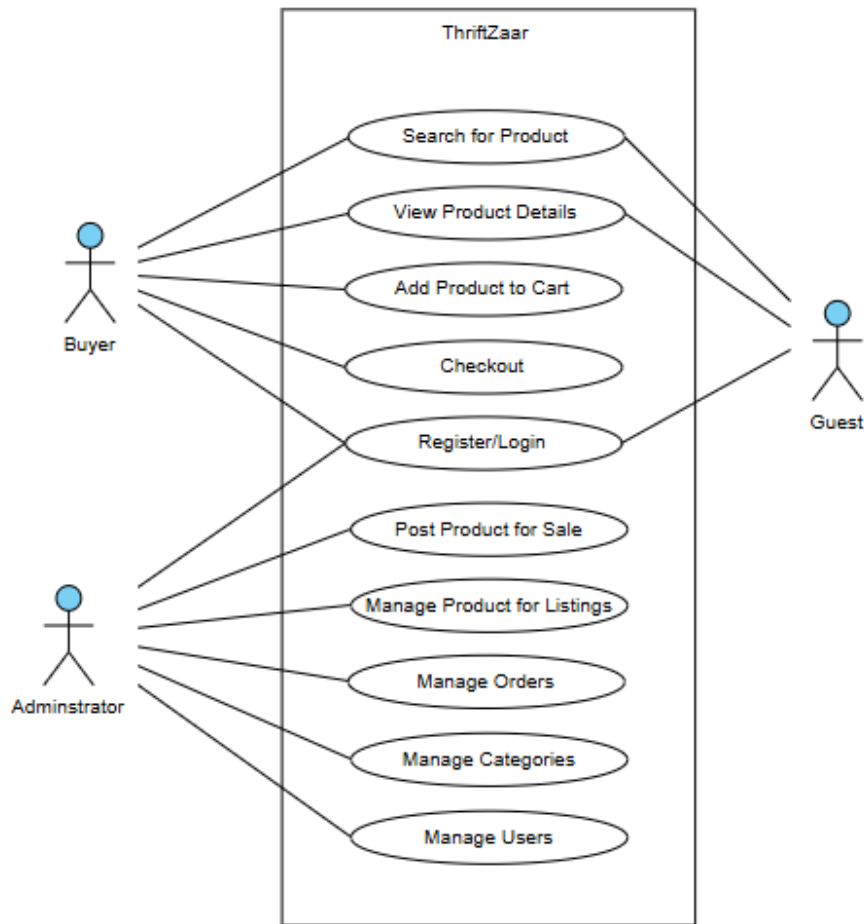


Figure 3.1.1: Use case diagram of ThriftZaar Nepal

ii. Non-Functional Requirements

- Usability Requirements
 - The system must provide a simple and user-friendly interface.
 - Navigation should be clear and easy to understand.
 - The website should be responsive across different devices.
- Performance Requirements
 - System response time should be fast during browsing and searching.
 - Pages should be loaded within acceptable time limits.
 - The system must support multiple users simultaneously.

- Security Requirements
 - Passwords must be encrypted in the database.
 - User authentication should be required for sensitive actions.
 - Admin access should be restricted through secure login.
- Reliability Requirements
 - The system should function consistently without frequent crashes.
 - Data integrity must be maintained during transactions.
- Availability Requirements
 - The system should be accessible online 24/7.
 - The system should work with multiple web browsers (Chrome, Mozilla, Opera, etc.).
- Maintainability Requirements
 - The code structure should be modular for easy updates.
 - Database design should allow future expansion.
 - Backup for databases should be available

3.1.2 Feasibility Analysis

Feasibility analysis is carried out to determine whether the proposed ThriftZaar Nepal system is practical and achievable within available resources, time, and technology. It helps evaluate whether the system can be successfully developed and implemented to meet the project objectives. The feasibility of the ThriftZaar Nepal project is analyzed under the following aspects:

- **Technical Feasibility**

Technical feasibility examines whether the necessary technologies, tools, and technical skills required for system development are available. [3] The ThriftZaar Nepal system is developed using standard web technologies such as HTML5, CSS3, JavaScript, PHP, Apache Server, and MySQL database, which are widely used and freely available. These technologies are compatible with most operating systems and do not require high-end hardware.

The development team possesses sufficient programming knowledge and experience to design and implement the system effectively. Development tools such as code editors, local servers, and web browsers are easily accessible. Since the application is web-based and runs on standard hosting environments, no specialized infrastructure is required. Therefore, the project is technically feasible.

- **Operational Feasibility**

Operational feasibility evaluates whether the system will function effectively in a real-world environment and whether users can easily operate it. The ThriftZaar Nepal platform is designed with a simple, user-friendly interface that allows users to browse clothing items, upload products, and manage listings with ease. The admin panel provides efficient management of products, categories, and users. Since the platform requires only basic computer and internet knowledge, users can operate it without extensive training. Hence, the system is operationally feasible.

- **Economic Feasibility**

Economic feasibility analyzes the financial viability of the project. The ThriftZaar Nepal system is developed using open-source technologies, which eliminates licensing costs. The project is carried out by students as part of academic coursework, reducing development expenses. The system requires only basic computing resources and low-cost web hosting for deployment. Maintenance and operational costs are minimal due to the simple design and widely supported technologies. Considering the low investment and potential benefits of providing an affordable online clothing marketplace, the project is economically feasible.

- **Schedule Feasibility**

Schedule feasibility determines whether the project can be completed within the allocated timeframe. The development of ThriftZaar Nepal follows a structured project plan that includes requirement analysis, system design, implementation, testing, and documentation phases. Tasks are distributed among team members to ensure efficient progress.

Since the system uses familiar technologies and has a well-defined scope, it can be completed within the academic project schedule. Therefore, the project is schedule feasible.

3.1.3 Data Modeling (ER-Diagram)

The ER diagram represents the conceptual data model of the ThriftZaar Nepal system. It shows the main database entities such as **Admin**, **Users**, **Clothes**, **Categories**, and **Orders**, along with their relationships. The diagram illustrates how data is structured, connected, and managed within the system. It helps in organizing the database design, maintaining data integrity, and supporting efficient system functionalities. [3]

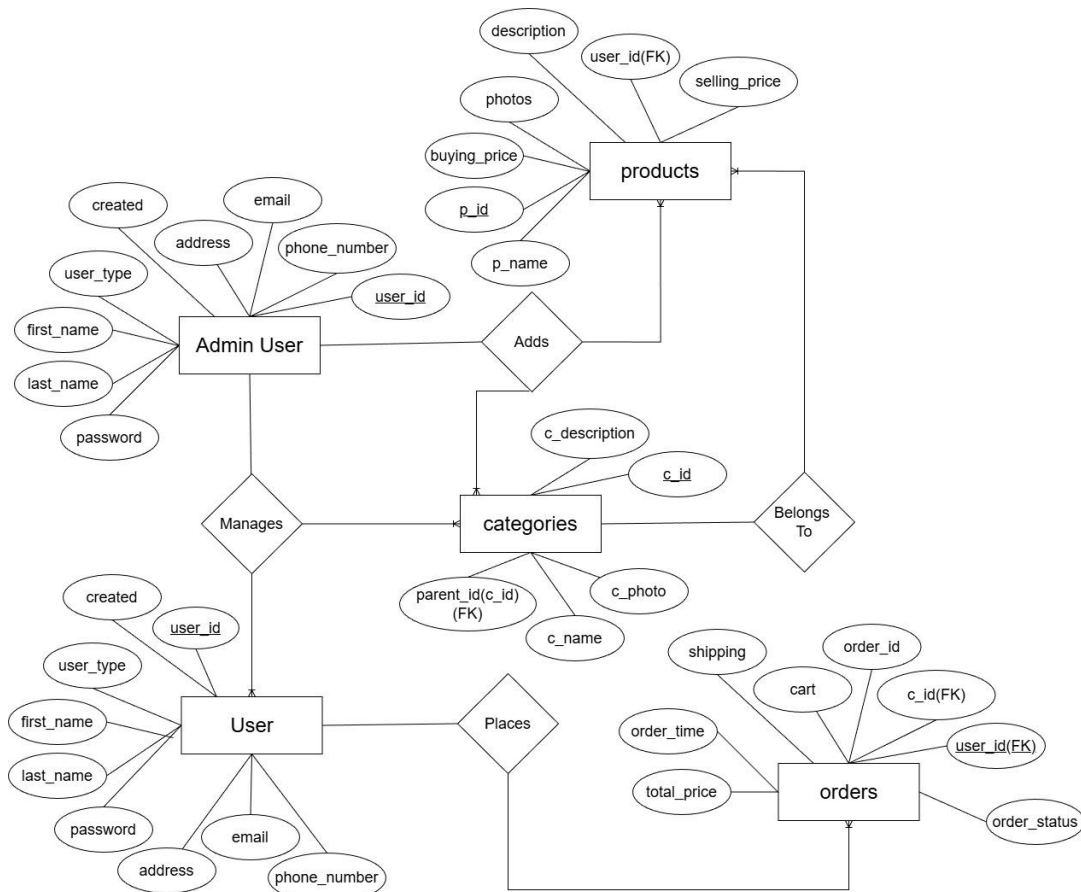


Figure 3.1.2: ER Diagram of ThriftZaar Nepal

3.1.4 Process Modeling (DFD)

Process modeling is used to represent how data moves through the ThriftZaar Nepal system and how different users interact with the system processes. Data Flow Diagrams (DFDs) illustrate the flow of information between users, system processes, and the database. The DFD helps in understanding system functionality and the interaction between various components.

The DFD for the ThriftZaar Nepal system is divided into two levels:

- DFD Level 0 (Context Diagram)
- DFD Level 1 (Detailed System Processes)

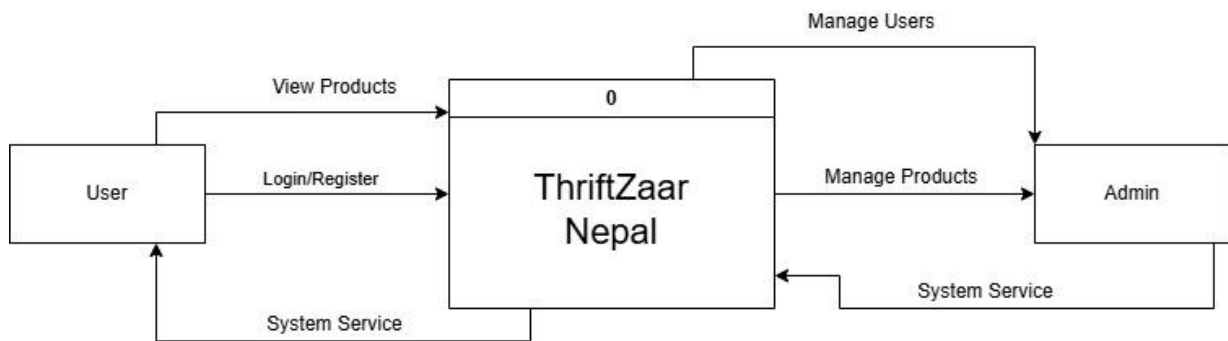


Figure 3.1.3: DFD-0

The above picture represents DFD level 0 of our application. The functionalities that can be done are roughly shown in the figure.

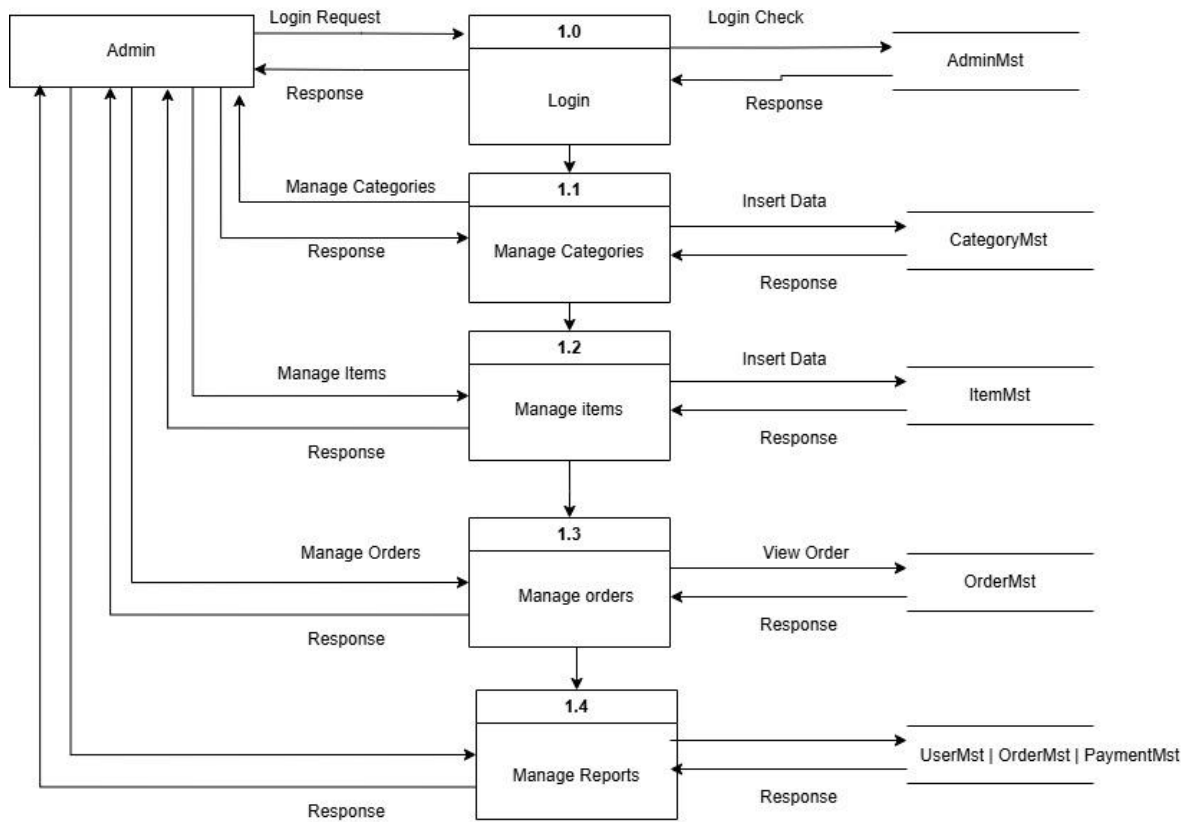


Figure 3.1.4: DFD-1

In the above DFD level 1 we can see all the operations that can be done by the admin.

3.2 System Design

3.2.1 Architectural Design

Architectural design defines the overall structure of the system and shows how different components interact with each other. [3] It provides a high-level view of the system, including modules, data flow, and communication between users, admin, and the database. The main purpose of architectural design is to ensure the system is scalable, maintainable, secure, and easy to manage.

For this **Second-Hand Clothing Web Application (ThriftZaar Nepal)**, **three-tier architecture** is used:

- **Presentation Layer (User Interface)**

This layer is responsible for interaction between the system and users (Admin, User, Guest). It includes:

- Home page
- Product listing page
- Login and signup page
- Admin dashboard
- Product upload and management interface
- Technologies: HTML, CSS, JavaScript, Bootstrap

- **Application Layer (Business Logic)**

This layer handles the main processing and system functionality. It performs:

- User authentication and authorization
- Product management
- Order and listing handling
- Validation of user input
- Communication between interface and database
- Technology: PHP

- **Data Layer (Database)**

This layer stores and manages system data. It includes:

- Admin Master Table
- User Table
- Product Table
- Category Table
- Technology: MySQL Database

Architecture Flow

Guest/User/Admin → Web Interface → Application Logic (PHP Server) → Database (MySQL)

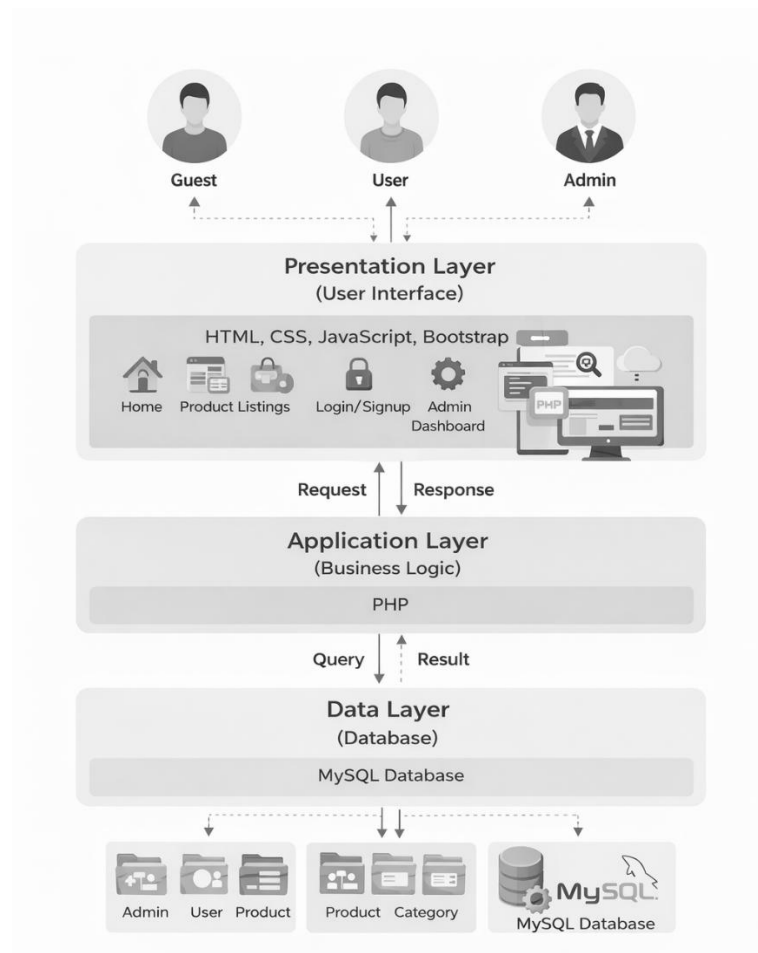


Figure 3.2.1: System Architectural Design

3.2.2 Database Schema Design

A database schema is the structure that represents the logical view of the entire database. It defines how the data is organized and how the relations among them are associated. It formulates all the constraints that are to be applied to the data.

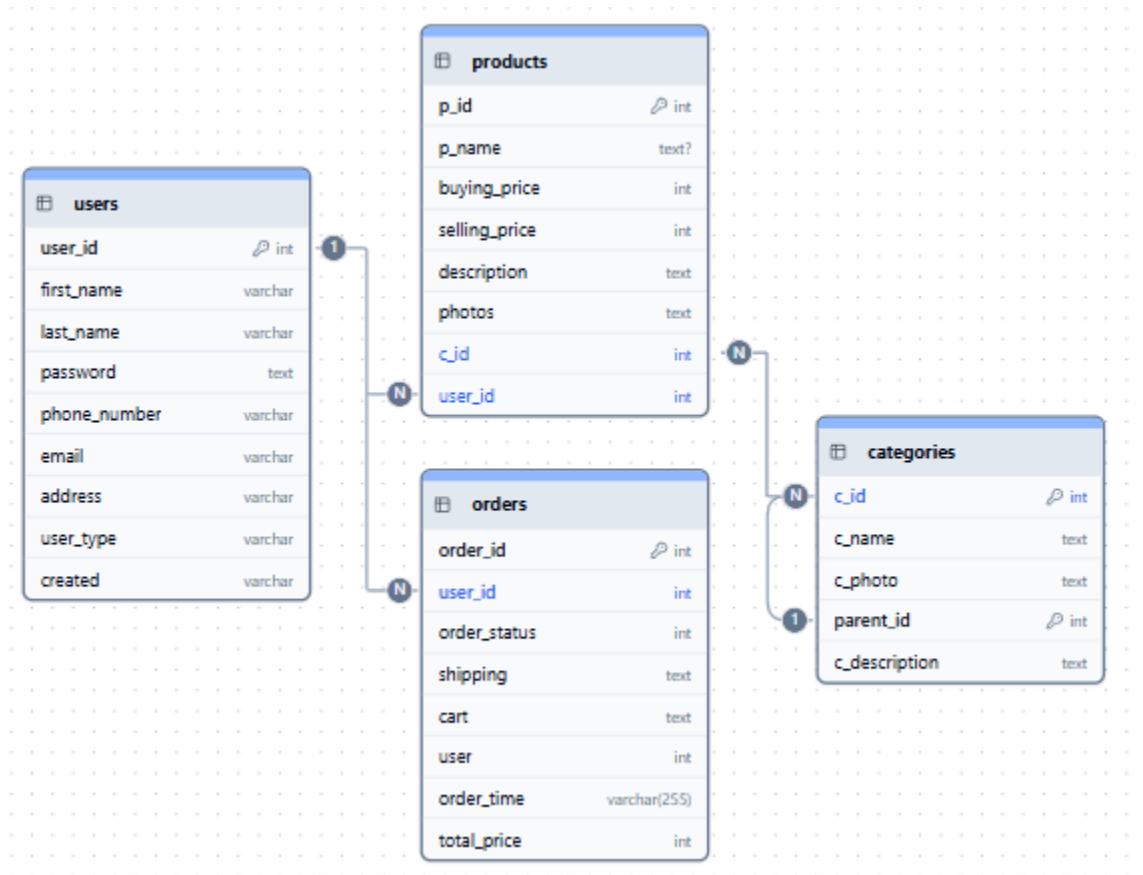


Figure 3.2.2: Database Schema Design

Here in the above figures show us the schema diagram of our application. It shows the connection between various tables.

3.2.3 Interface Design (UI Interface / Interface Structure Diagrams)

Before implementing the actual design of the project, a few user interface designs are constructed to visualize the user interaction with the system as they browse for clothes, search, register, login and purchase. The main objective of the interface design is to provide a simple, user-friendly, and efficient environment where users and administrators can easily interact with the system. The interfaces are designed to be responsive, accessible, and easy to navigate for different types of users including Guests, Registered Users, and Administrators.

The interface structure is divided into different modules based on user roles and system functionalities.

A. Home Page Interface

The home page serves as the main entry point for all users. It provides an overview of available clothing items and general system navigation.

Main Features:

- Navigation bar (Home, Products, Login, Register)
- Featured clothing products
- Footer with contact information

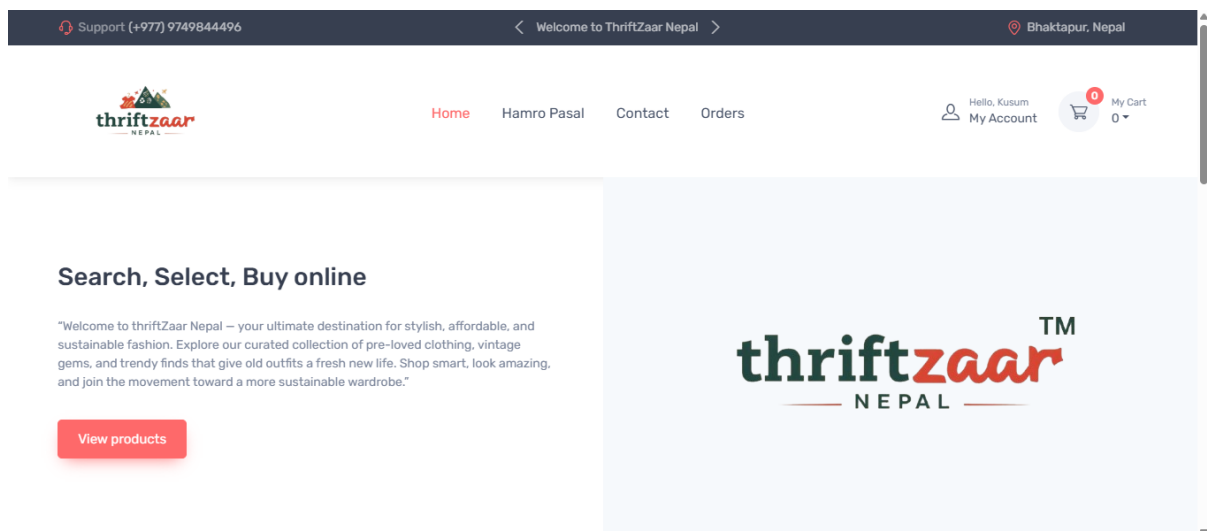


Figure 3.2.3.1: Home Page Interface

B. Login and Registration Interface

This interface allows users and administrators to authenticate themselves into the system.

Login Interface Features:

- Email input
- Password input
- Login button

Registration Interface Features:

- Name input
- Email input
- Password creation
- Contact information
- Submit button

The image displays a user authentication and registration interface. It is divided into two main sections: 'Sign in' and 'No account? Sign up'.

Sign in section:

- Header: 'Sign in'
- Option: 'With social account:' followed by icons for Google, Facebook, and Twitter.
- Section: 'Or using form below'
- Form fields: 'Email' (with an envelope icon) and 'Password' (with a lock icon and a toggle eye icon).
- Checkbox: 'Remember me' with a checked red box.
- Link: 'Forgot password?' in red text.
- Button: 'Sign In' in a red box with a white arrow icon.

No account? Sign up section:

- Header: 'No account? Sign up'
- Text: 'Registration takes less than a minute but gives you full control over your orders.'
- Form fields: 'First Name', 'Last Name', 'E-mail Address', 'Phone Number', 'Password', and 'Confirm Password'.
- Button: 'Sign Up' in a red box with a white person icon.

Figure 3.2.3.2: Login and Registration Interface

C. Product Listing Interface

This interface displays available clothing items to users in a structured layout.

Features:

- Product image
- Product name
- Price display
- Category filtering
- Search functionality
- View details option

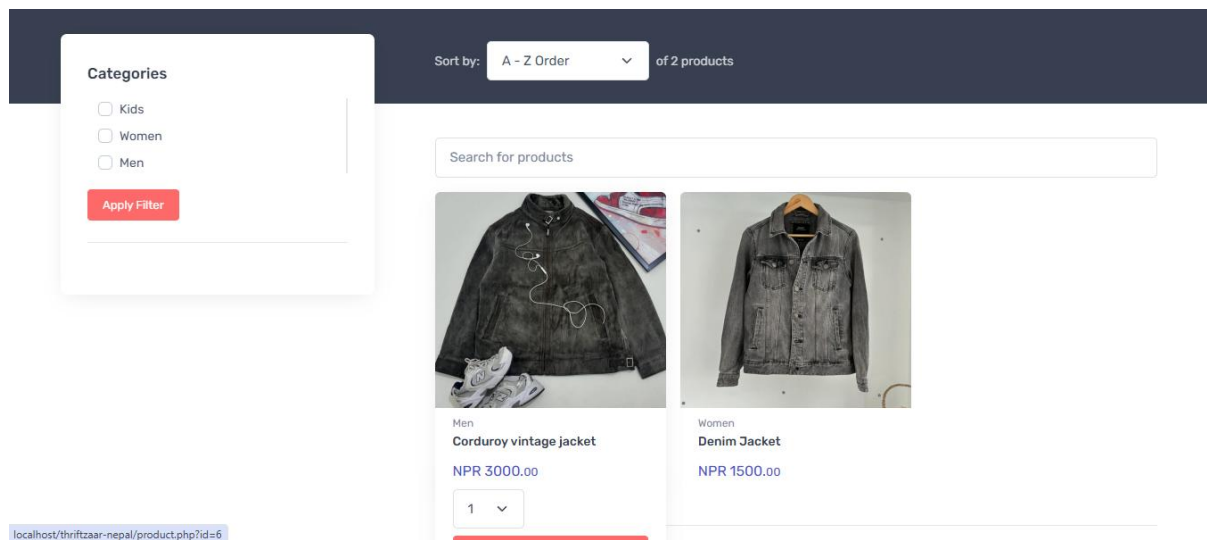


Figure 3.2.3.3: Product Listing Interface

D. Product Detail Interface

This interface provides detailed information about a selected clothing item.

Features:

- Product images
- Description
- Price
- Availability status
- Add to cart option

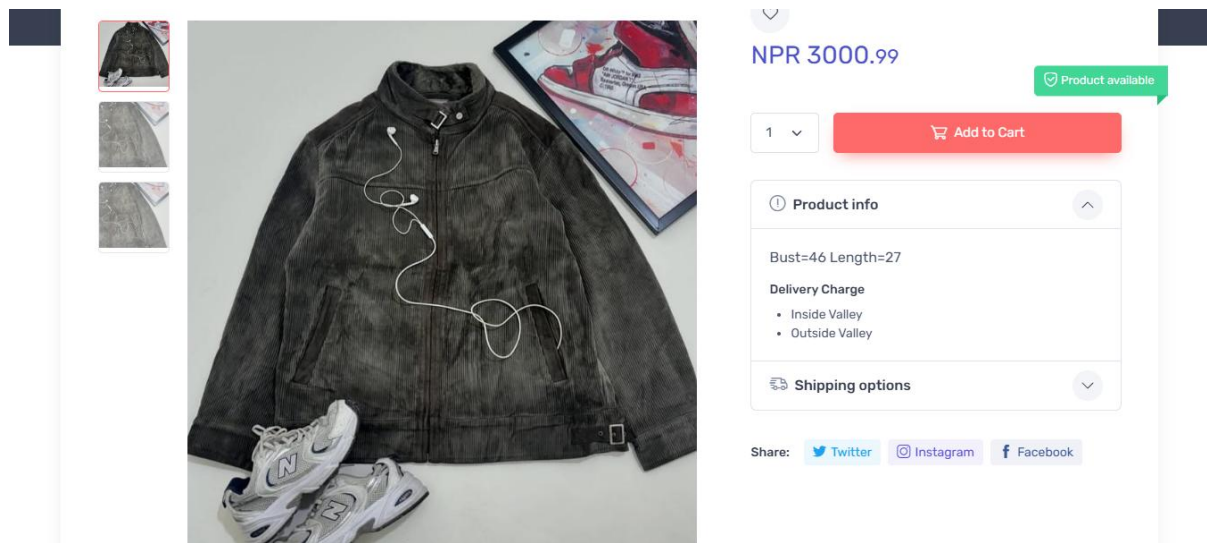


Figure 3.2.3.4: Product Detail Interface

E. User Dashboard Interface

The user dashboard allows registered users to manage their account and purchases.

Features:

- Profile management
- Order history
- Account settings
- Logout option

The screenshot displays the 'Profile info' page of a user dashboard. The page has a dark blue header with the title 'Profile info' on the left and a breadcrumb trail 'Home > Account > Profile info' on the right. A 'Sign out' button is located in the top right corner. On the left side, there is a user profile card for 'Kusum Darlami' with the email 'darlamikusum040@gmail.com'. The card includes a 'Good Day' greeting, a profile picture, and a sidebar menu with 'My Account', 'Profile info' (highlighted), and 'My Orders' (showing 0 orders). The main content area is titled 'Update your profile details below:' and contains several form fields: 'First Name' (Kusum), 'Last Name' (Darlami), 'Email Address' (darlamikusum040@gmail.com), 'Phone Number' (9749844496), 'New Password', and 'Confirm Password'. There are eye icons for password visibility. An 'Update profile' button is at the bottom.

Profile info

Home > Account > Profile info

Sign out

Good Day

Kusum Darlami
darlamikusum040@gmail.com

My Account

Profile info

My Orders 0

Update your profile details below:

First Name Last Name

Kusum Darlami

Email Address Phone Number

darlamikusum040@gmail.com 9749844496

New Password Confirm Password

Update profile

localhost/thriftzaar-nepal/account-profile.php

Figure 3.2.3.5: User Dashboard Interface

F. Administrator Dashboard Interface

The administrator dashboard provides system control and management functionality.

Features:

- Manage users
- Add/Edit/Delete products
- Manage categories
- Monitor orders
- System management panel

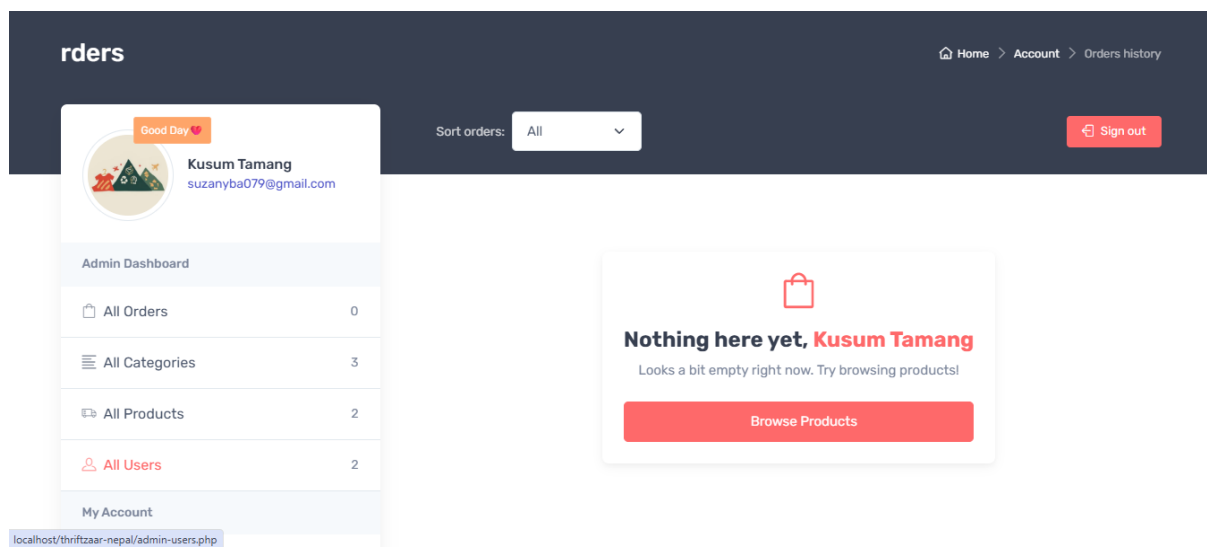


Figure 3.2.3.6: Admin Dashboard Interface

3.2.4 Physical DFD

The **Physical Data Flow Diagram (DFD)** of ThriftZaar Nepal illustrates the actual implementation of system processes, user interactions, and database operations. It shows how users and administrators interact with the system through physical components such as web forms, PHP scripts, and MySQL databases.

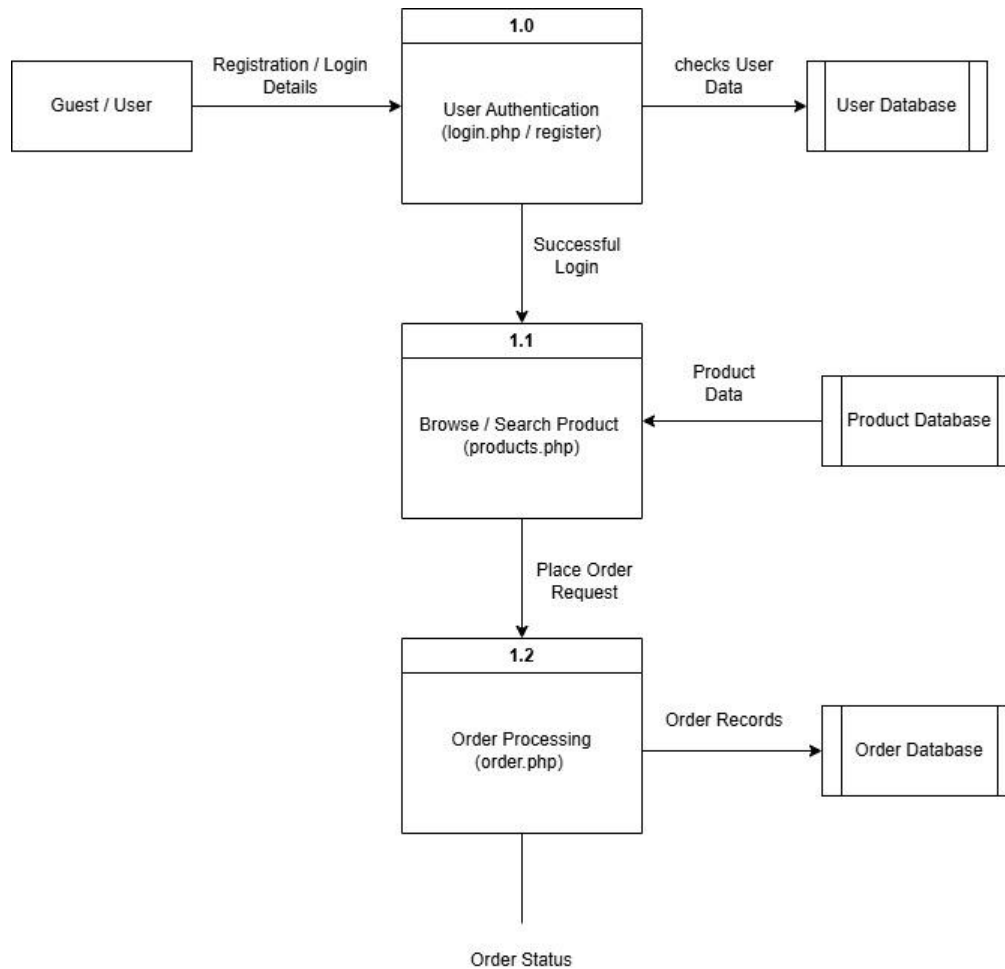


Figure 3.2.4.1: Physical DFD of ThriftZaar Nepal (User)

Users and guests can register, log in, browse products, search for clothing items, and place orders through the web interface. All authentication and data processing are handled by server-side PHP scripts, and data is stored and retrieved from MySQL database tables such as User, Product, and Order.

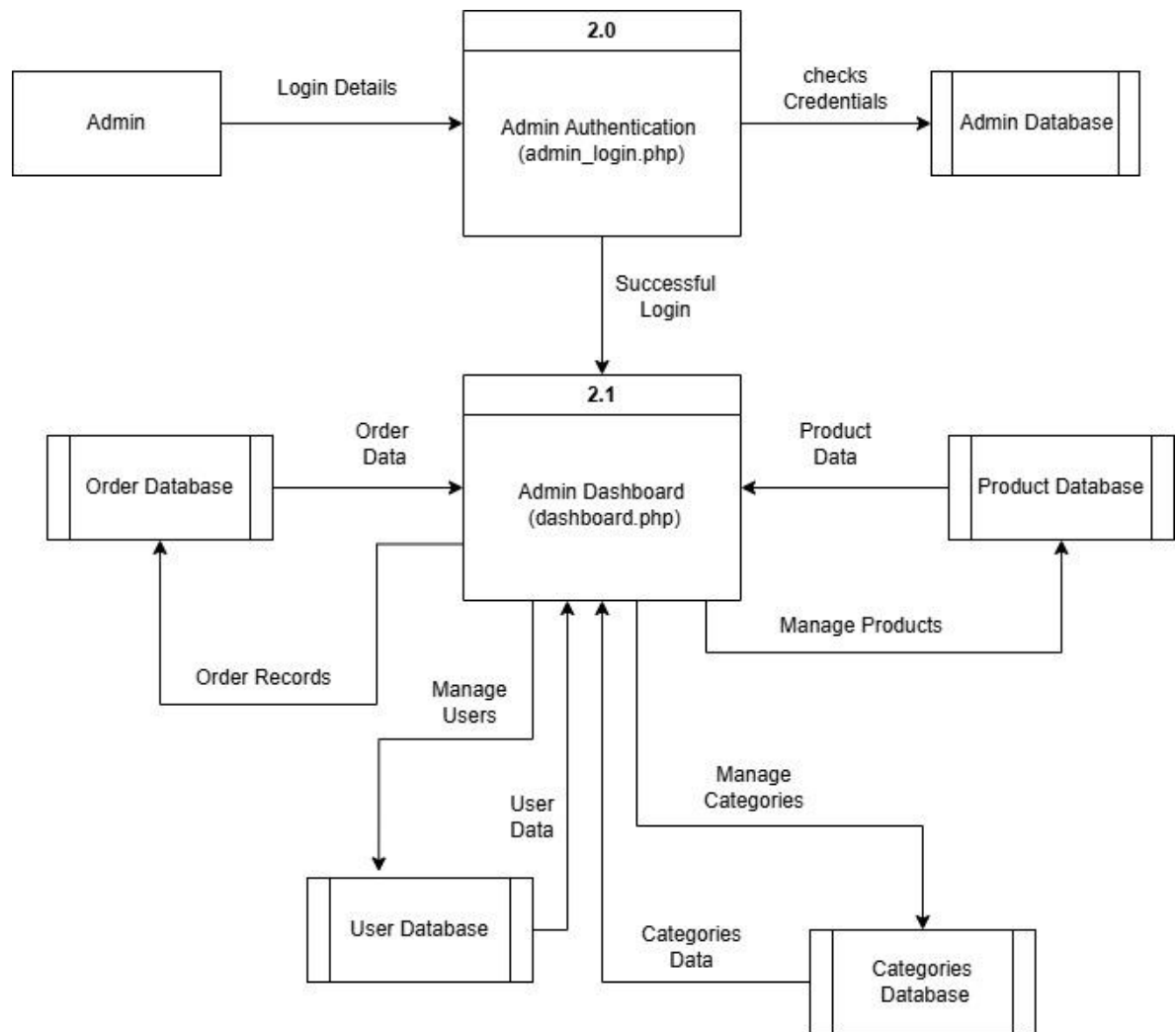


Figure 3.2.4.2: Physical DFD of ThriftZaar Nepal (Admin)

The administrator logs into the admin dashboard to manage users, products, categories, and orders. Admin actions such as adding, updating, or deleting records directly interact with the database through secured backend processes.

This physical DFD provides a clear understanding of how data flows through the system in real-time and how the system operates in a practical environment.

Chapter 4: Implementation and Testing

4.1 Implementation

After completing the design phase of the system, the next step is implementation so that the system can be deployed according to user requirements and satisfaction. The implementation phase involves various technical processes, resources, and detailed procedures. Since it is not possible to include every technical detail within the scope of this report, only the major aspects and key components of the system implementation are discussed in the following sections.

4.1.1 Tools Used

The development of ThriftZaar Nepal required a combination of programming languages, database platforms, and software tools. These tools were carefully chosen to ensure efficiency, compatibility, and maintainability:

- **CASE Tools:** CASE (Computer-Aided Software Engineering) tools like draw.io, were used for system modeling, diagramming, and documentation. These tools helped in maintaining clear workflows and understanding relationships between different modules.
- **Programming Languages:**
HTML5, CSS3, JavaScript, Bootstrap are used to create a responsive and interactive user interface that is compatible with multiple devices. PHP is used for implementing the business logic and server-side operations such as authentication, order processing, and product management.
- **Database Platform:** MySQL is used to store and manage all system data, including user accounts, product listings, orders, and categories.
- **Development Environment** Local servers such as XAMPP and LAMP, along with code editors like VS Code, were used to write, test, and debug code efficiently.

These tools provided a robust foundation for building a system that is scalable, secure, and easy to maintain.

4.1.2 Implementation Details of Modules

The ThriftZaar Nepal system is divided into several functional modules, each designed to handle a specific set of tasks. The implementation of each module was carried out using PHP scripts that interact with the MySQL database and the front-end interface. The major modules are described below:

- **Admin Module:** The Admin module is responsible for managing the entire platform. Administrators can manage users, products, categories, and orders. The key procedures include:
 - **admin-product-add.php** - Adds new products to the system by collecting product details from the admin interface, validating inputs, and storing them in the database.
 - **admin-user.php** - Allows the admin to view, delete, or assign roles to users.
 - **admin-product-delete.php** - Deletes invalid or outdated products from the database.
 - **admin-product-edit.php** - Updates existing product information, ensuring data consistency.
 - **admin-categories-add.php** – Adds new categories to the system.
- **User Module:** Registered users can browse, search, and purchase products. Important scripts include:
 - **cart-process-add.php** - Adds selected products to the user's cart, verifying quantity and availability before updating the cart table.
 - **login.php** - Authenticates users before granting access to their dashboard.
 - **register.php** - Handles new user registration.
 - **complete-order.php** - Processes orders and updates the orders and inventory tables.
- **Guest Module:** Guests can browse products and view details without registering. Key scripts include:
 - **hamro-pasal.php** - Displays available clothing items on the product page and allows searching and filtering of products.
 - **product.php** - Provides detailed information about selected products.

4.2 Testing

Testing is an integral part of system development. It ensures that the system functions correctly, meets the specified requirements, and provides a seamless experience to users. The testing of ThriftZaar Nepal was carried out in multiple stages, including unit testing and system testing.

4.2.1 Test Cases for Unit Testing

Unit testing focuses on verifying that each individual module works as intended. By testing functions and procedures independently, errors can be identified and corrected at an early stage. We both acted as a tester and tested most of the major and minor functionalities of web app and recorded the outputs on the tables below:

Table 1: Test Case For User Login

Test Id	Description	Test Data	Expected Output	Actual Output	Result
TC_01	User enters a wrong password	Email:suzanyba079@gmail.com Password: pass123	Display : “Login Failed! You entered wrong email or password.’	As Expected ,	Pass
TC_02	User enters correct password	Email:suzanyba079@gmail.com Password: admin	Logged into user profile page	As Expected ,	Pass

Table 2: Test Case For User Registration

Test Id	Description	Test Data	Expected Output	Actual Output	Result
TC_03	User tries to register into the system with registered email.	First Name : Sujan Last Name: Tamang Email:suzanyba079@gmail.com Phone : 9762447050 Password: pass1 Confirm Password:pass1	Display : “You already have an account with this email address!”	As Expected ,	Pass
TC_04	User register with new email.	First Name : Kusum Last Name: Darlami Email:darlamikusum@gmail.com Phone : 9749844496 Password: pass1 Confirm Password:pass1	Logged into user profile page	As Expected ,	Pass

Table 3: Test Case For Adding and Searching Products

Test Id	Description	Test Data	Expected Output	Actual Output	Result
TC_05	Admin tries to add a new product	Product Name : Blazer Product Category: Women Photos: blazer.jpg Buying Price: 1000 Selling Price: 1500 Description: Good Quality Blazer, Size: 37, length: 27	Display : “Product Added Success Fully’ and go to admin product page	As Expected ,	Pass
TC_06	User Searching for a product	Denim Jacket product is in the shop, and user search key is denim	Matching products displayed	As Expected ,	Pass

Table 4: Test Case For Adding Category

Test Id	Description	Test Data	Expected Output	Actual Output	Result
TC_07	Adding the new category by admin	Category Name: Sweatshirt Photo: sweatshirt.jpg Parent Category: Men Description: Men good quality sweat shirts	Displays: “Category added successfully” and redirect to all categories page	As Expected ,	Pass

4.2.2 Test Cases for System Testing

System testing evaluates the integrated system as a whole. It ensures that all modules interact correctly and that the system meets both functional and non-functional requirements. Example test scenarios include:

Table 5: Test Cases For System Testing

Test Id	Description	Test Data	Expected Output	Actual Output	Result
TC_08	Verify login functionality for registered users	Email:darlamikusum@gmail.com Password: user	User dashboard displayed	As Expected ,	Pass
TC_09	Verifying updating the user information	Changing phone number.	Display: "Profile Updated successfully" and Database updated accordingly	As Expected ,	Pass
TC_10	Verify access control for registered pages	Trying to access account profile page without logging in.	Access denied	As Expected ,	Pass
TC_11	Test order placement for a clothing item	Product Id, Quantity, shipping details.	Order recorded; inventory updated	As Expected ,	Pass

Chapter 5: Conclusion and Future Recommendations

5.1. Lesson Learnt / Outcome

The development of the ThriftZaar Nepal system provided valuable practical experience in applying software engineering concepts to a real-world web application project. Throughout the project lifecycle, starting from requirement analysis to system implementation and testing, the development we gained a deeper understanding of structured system development processes.

One of the major outcomes of this project was the improvement of technical skills in web development technologies such as HTML, CSS, JavaScript, PHP, and MySQL. We learned how to design user-friendly interfaces, implement secure authentication systems, and manage relational databases effectively. Additionally, knowledge of system modeling techniques such as ER diagrams, DFDs, and architectural design was strengthened during the planning and design phases.

The project also enhanced teamwork and communication skills. Collaborating on tasks such as requirement gathering, system design, coding, debugging, and testing helped the team understand the importance of coordination and proper task distribution. Problem-solving abilities were improved by identifying system errors, debugging code, and optimizing performance. Overall, this project provided hands-on experience in building a complete web-based application and strengthened both technical and project management skills.

5.2 Conclusion

The ThriftZaar Nepal system was successfully designed and developed as a second-hand clothing web application that connects buyers and sellers through an easy-to-use online platform. The system meets the functional and non-functional requirements defined during the analysis phase and provides a secure, responsive, and user-friendly environment for users, guests, and administrators.

The application allows users to browse clothing items, search for products, manage accounts, and place orders efficiently. The administrator has full control over product listings, categories, and user management through a dedicated dashboard. The implementation of a three-tier architecture ensures that the system is scalable, maintainable, and organized.

Testing results confirmed that the system performs reliably and meets performance, usability, and security requirements. The use of open-source technologies helped in minimizing development costs while ensuring compatibility across different platforms and devices.

In conclusion, ThriftZaar Nepal demonstrates the successful application of system analysis, design principles, and modern web technologies to develop a functional and practical web-based marketplace solution.

5.3 Future Recommendations

Although the current version of ThriftZaar Nepal fulfills its primary objectives, several improvements and additional features can be implemented in future versions to enhance system functionality and user experience. Some recommendations include:

- **Online Payment Integration:** Implement secure digital payment systems such as eSewa or Khalti to allow direct online transactions.
- **Mobile Application Development:** Develop a mobile application to provide easier access for users on smartphones.
- **Advanced Search and Recommendation System:** Introduce AI-based product recommendations and improved filtering options.
- **User Review and Rating System:** Allow users to review and rate products and sellers to improve trust and transparency.
- **Real-Time Notifications:** Implement email or SMS notifications for order updates and account activities.
- **Performance Optimization:** Improve system performance through caching techniques and optimized database queries.
- **Multi-Vendor Support:** Allow multiple sellers to manage their own products independently within the platform.

These future improvements can make ThriftZaar Nepal more scalable, secure, and competitive in the online clothing marketplace environment.

Appendices

Appendix A: Screen Shots

This section includes important screenshots of the system interfaces that visually represent the functionality and usability of the application. The screenshots are captured during system execution to show real-time operation of different modules.

Admin Dashboard

All Orders

3

All Categories

5

All Products

6

All Users

1

My Account

Profile info

Products

6

Image	Product Name	Selling Price	Buying Price	Description	Actions
	Corduroy vintage jacket	3000	2500	Bust=46 Length=27	
	Denim Jacket	1500	1000	Bust=38 Length=25	
	Blazer	1500	1000	Bust=42, Length=30	
	Teenie Weenie Sweatshirt	1300	1000	Bust=48, length=27	
	Cardigan	1300	1000	Bust=36, length=17	

Figure A.1: Product Listing Page

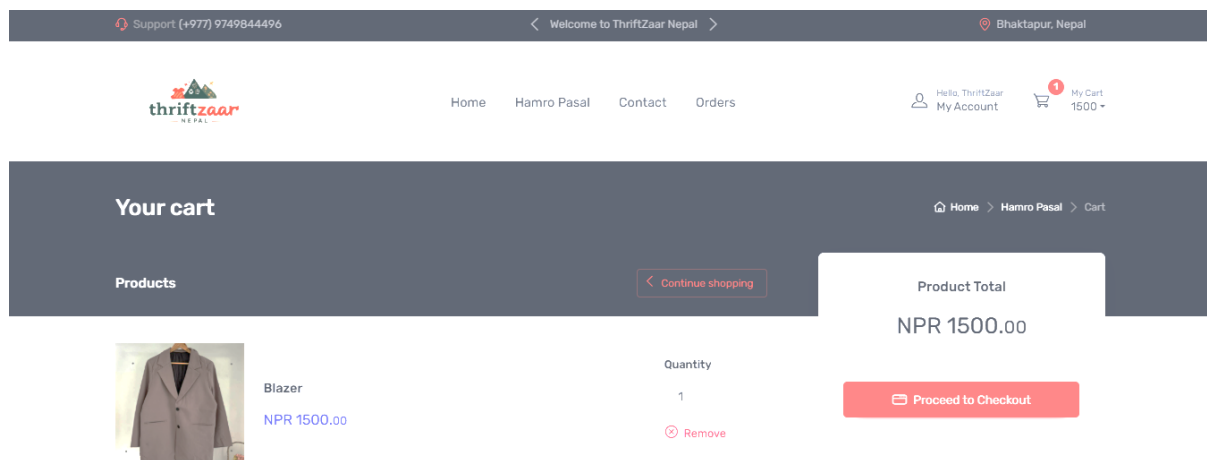


Figure A.2: Cart Page

Orders

[Home](#) > [Account](#) > [Orders history](#)

Reward points

Good Day

ThriftyZaar Admin

admin@gmail.com

Admin Dashboard

All Orders

3

All Categories

5

All Products

6

All Users

1

My Account

Profile info

Sort orders:

All

Sign out

Order #	Date Purchased	Status	Total	Action
11	January 01, 1970	Cancelled	NPR 3,000	—
12	January 01, 1970	Completed	NPR 1,300	—
13	January 01, 1970	In Progress	NPR 1,200	Cancel Order

Figure A.3: Admin Order Listing Page

Checkout

[Home](#) > [Hamro Paisal](#) > [Checkout](#)

1

2

3

Cart

Checkout

Review

384

ThriftyZaar Admin

admin@gmail.com

Edit profile

Shipping address

First Name

ThriftyZaar

Last Name

Admin

E-mail Address

admin@gmail.com

Phone Number

9762447050

Address

Bhaktapur,Nepal

Order summary

Blazer

NPR 1500.00 x 1

Subtotal:

NPR 1500.00

Delivery Charge:

NPR 100.00

NPR 1600.00

Figure A.4: Checkout Page

37

Figure A.5: User Registration and Login Page

Figure A.6: Client-Side Validation

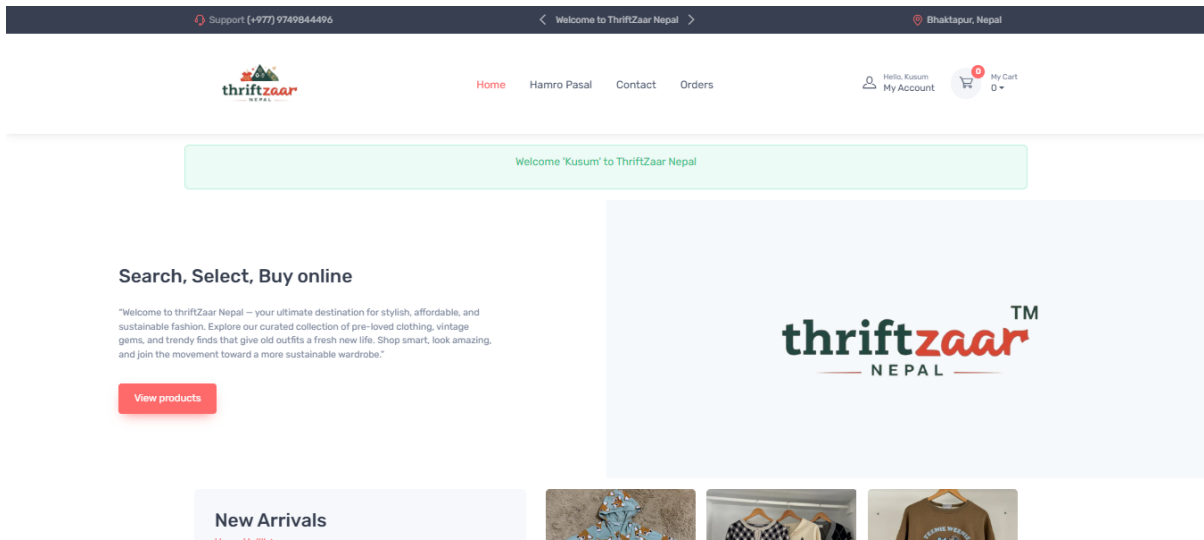


Figure A.7: Successful Registration

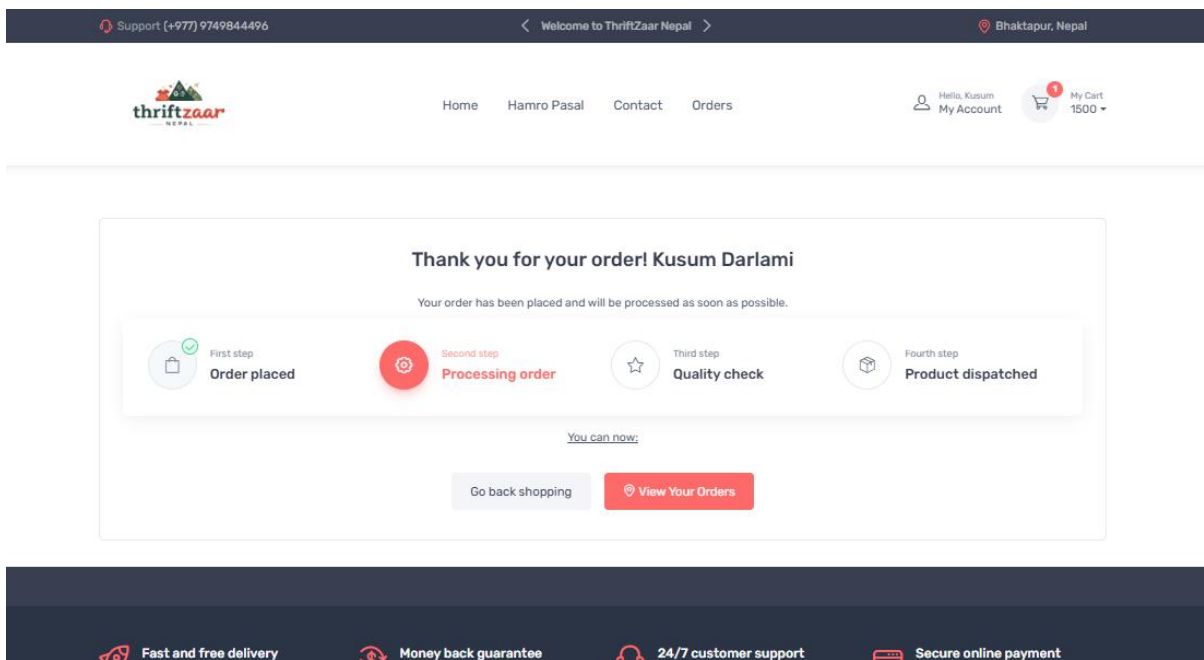


Figure A.8: Order Completion Page

Appendix B: Source Codes

This section contains selected and important source code files used in the development of the ThriftZaar Nepal system. The codes demonstrate how different modules of the system were implemented using PHP, MySQL, HTML, CSS, and JavaScript.

```
1  if (session_status() === PHP_SESSION_NONE) {
2      session_start();
3  }
4
5  define("BASE_URL", "http://localhost/thriftzaar-nepal");
6
7  $conn = mysqli_connect("localhost", "root", "", "thriftzaar");
8
9
```

Figure B.1: Database Configuration

```
1  <?php
2  require_once("../modules/config.php");
3
4  if (isset($_POST['id']) && isset($_POST['table'])) {
5
6      $id = intval($_POST['id']);
7      $table = $_POST['table'];
8
9
10     if ($table == "products") {
11         $sql = "DELETE FROM $table WHERE p_id = $id";
12     } elseif ($table == "categories") {
13         $sql = "DELETE FROM $table WHERE c_id = $id";
14     } elseif ($table == "users") {
15         $sql = "DELETE FROM $table WHERE user_id = $id";
16     } else {
17         echo "invalid";
18         exit;
19     }
20
21     if (mysqli_query($conn, $sql)) {
22         echo "success";
23     } else {
24         echo mysqli_error($conn);
25     }
26
27 }
28 ?>
```

Figure B.2: admin-remove.php

```

1  <?php
2
3  require_once("modules/config.php");
4
5
6  $email = trim($_POST['email']);
7  $password = trim($_POST['password']);
8  $password_1 = trim($_POST['password_1']);
9  $phone_number = trim($_POST['phone']);
10 $last_name = trim($_POST['last_name']);
11 $first_name = trim($_POST['first_name']);
12
13 $sql = "SELECT * FROM users WHERE email = '{$email}'";
14 $res = $conn->query($sql);
15
16 if($res->num_rows>0){
17     alert("danger","You already have an account with this email address!!!");
18     header("Location: login.php");
19     die();
20 }
21
22 //encrypts the password
23 $plain_password = $password;
24 $password = password_hash($password, PASSWORD_DEFAULT);
25 $created = time();
26
27 $sql = "INSERT INTO users (
28     first_name,
29     last_name,
30     phone_number,
31     password,
32     email,
33     user_type,
34     created
35 ) VALUES (
36     '{$first_name}',
37     '{$last_name}',
38     '{$phone_number}',
39     '{$password}',
40     '{$email}',
41     'customer',
42     '{$created}'
43 )";
44
45 if($conn->query($sql)){
46     login_user($email,$plain_password);
47     alert("success","Welcome '{$first_name}' to ThriftZaar Nepal");
48     header('Location: /thriftzaar-nepal');
49 }else{
50     alert("danger","Failed, Something went wrong!");
51     header("Location: login.php");
52     die();
53 }

```

Figure B.3: register-logic.php

```

1  <?php
2  require_once("modules/config.php");
3
4  $id = isset($_GET["id"]) ? (int)$_GET["id"] : 0;
5
6  if (isset($_SESSION['cart'])) {
7
8      foreach ($_SESSION['cart'] as $key => $value) {
9
10         if ($value['p_id'] == $id) {
11             unset($_SESSION['cart'][$key]);
12             echo "success";
13             alert("danger", "Removed To Cart Successfully");
14             exit;
15         }
16     }
17 }
18
19 echo "error";
20

```

Figure B.4: cart-process-remove.php

```

1  <?php
2  require_once("../modules/config.php");
3  protected_area();
4
5
6  if ($_SERVER["REQUEST_METHOD"] == "POST") {
7
8      if (!isset($_POST['order_id'])) {
9          header("Location: my-orders.php");
10         exit();
11     }
12
13     $order_id = intval($_POST['order_id']);
14     $user_id = $_SESSION['user_id'];
15     // Update order status to Canceled (-1)
16     $sql = "UPDATE orders SET order_status = -1 WHERE order_id = $order_id";
17     global $conn;
18     $conn->query($sql);
19
20
21     header("Location: account-orders.php");
22     exit();
23 }
24 ?>

```

Figure B.5: cancel-order.php

References

- [1] "depop," [Online]. Available: <https://www.depop.com>.
- [2] "Thriftstore Nepal," [Online]. Available: <https://www.thriftstorenepal.com>.
- [3] M. G. Ramesh Singh Saud, "Software Engineering," KEC Publication and Distribution Pvt.Ltd, 2077.